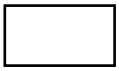
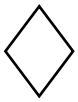
12. Entity Relationship Diagram

This section covers the theoretical description of the Entity-Relationship design of the Medical Appointment & Prescription System, including a description of the notations used, database entities, attributes, and cardinalities.

12.1 ER Components & Symbols

Symbol / Element Name	Visual Representation	Description
Entity		Represents a core database table or real-world object within the system(e.g., USERS, PATIENTS, DOCTORS).
Relationship		Represents the operational association or interaction linking two entities together(e.g., Has Profile, Books, Attends).
Attribute		Represents the specific properties, data fields, or columns belonging to an entity(e.g., username, age, specialization).
Primary key		Identifies the unique attribute used to uniquely distinguish each record within an entity table(e.g., user_id, patient_id).
One to one	1:1	Defines a strict mapping constraint where a single record in one entity relates to exactly one record in another(e.g., Users to Patients).
One to many	1:N	Defines a mapping constraint where a single record in an entity relates to multiple records in another(e.g., Patients to Appointments).

12.2 Entity and Attributes

i. Users

- **Purpose:** Stores the data related to central authentication such as usernames, the secure hashed passwords, and roles of all the users of the system.
- **Attributes**
  - user_id (Primary Key, INT)
  - username (VARCHAR)
  - password (VARCHAR)
  - role (VARCHAR)

iiPatients

- **Purpose:** Stores the contact and personal information of all the people eligible to receive services through the system.
- **Attributes:**

  - patient_id (Primary Key, INT)
  - full_name (VARCHAR)
  - age (INT)
  - gender (VARCHAR)
  - contact_number (VARCHAR)
  - residential_address (TEXT)
  - blood_group (VARCHAR)
  - registration_date (TIMESTAMP)

i. Doctors

- **Purpose:** Stores the personal and professional information of all the medical personnel such as their specializations, contact information, and schedule.
- **Attributes:**

  - doctor_id (Primary Key, INT)
  - full_name (VARCHAR)
  - specialization (VARCHAR)
  - contact_number (VARCHAR)
  - working_hours (VARCHAR)
  - consultation_fee(DECIMAL)
  - availability_status (BOOLEAN / VARCHAR)

iv. Appointments

- **Purpose:** Stores the schedule of doctors' appointments including the date and time of each appointment
- **Attributes:**

  - appointment_id (Primary Key, INT)
  - appointment_date (DATE)
  - appointment_time (TIME)
  - status (VARCHAR)
  - created_at (TIMESTAMP)

v. Prescriptions

- **Purpose**: Stores the medical treatment records of all the patients in electronic form including the prescribed drugs and notes made by doctors during each visit.
- **Attributes:**

  - prescription_id (Primary Key, INT)
  - issue_date (DATE)
  - medicines_and_dosage (TEXT)
  - clinical_notes (TEXT)

12.3 Relationship

Entity one	Relation	Entity two	Relation Linkage
Users	register	Patients	The Patients table has a foreign key

			(registered_by) that points to the primary key (user_id) of the Users table, indicating who among the staff or admin created a patient record..
Users	has profile	Doctors	The Doctors table contains a foreign key (user_id) linked to the primary key (user_id) in the Users table.
Patients	books	Appointments	The Appointments table contains a foreign key (patient_id) linked to the primary key (patient_id) in the Patients table
Doctors	attends	Appointments	.The Appointments table contains a foreign key (doctor_id) linked to the primary key (doctor_id) in the Doctors table.
Appointments	generates	Prescriptions	The Prescriptions table contains a foreign key (Appointment_id) linked to the primary key (appointment_id) in the Appointments
Patients	receives	Prescriptions	The Prescriptions table contains a foreign key (patient_id) linked to the primary key (patient_id) in the Patients table.
Doctors	issues	Prescriptions	The Prescriptions table contains a foreign key (doctor_id) linked to the primary key (doctor_id) in the Doctors table.

12.4 Cardinality Mapping

Entity One	**Cardinality**	**Entity Two**	**Mechanism**
Users	1:N	Patients	A single staff or administrative user can register multiple patient profiles over time (registered_by foreign key linkage)
Users	1:1	Doctors	Each professional doctor’s data profile is tied to a singular administrative user’s account login.

Patients	1:N	Appointments	Every single registered patient is allowed and able to make multiple appointment slots for consultations.
Doctors	1:N	Appointments	A single medical doctor arranges and schedules multiple appointment slots for consultations within their working hours.
Appointments	1:1	Prescriptions	Every finished appointment slot for consultation generates a singular digital prescription record.
Patients	1:N	Prescriptions	A single patient holds multiple historical prescription records, relating to their medical history.
Doctors	1:N	Prescriptions	A medical doctor issues multiple prescription records to multiple patients, within their working hours.

12.5 Diagram

